European Journal of Pharmacology 615 (2009) 133–138

Contents lists available at ScienceDirect

European Journal of Pharmacology

journal homepage: www.elsevier.com/locate/ejphar

Cardiovascular Pharmacology

# Spinal sumatriptan inhibits capsaicin-induced canine external carotid vasodilatation via 5-$HT_{1B}$ rather than 5-$HT_{1D}$ receptors

Enriqueta Muñoz-Islas [a], Jair Lozano-Cuenca [a], Abimael González-Hernández [a], Martha B. Ramírez-Rosas [a], Araceli Sánchez-López [a], David Centurión [a], Antoinette MaassenVanDenBrink [b], Carlos M. Villalón [a,*]

[a] Departamento de Farmacobiología, Cinvestav-Coapa, Czda. de los Tenorios 235, Col. Granjas-Coapa, Deleg. Tlalpan, 14330 México D.F., México
[b] Division of Pharmacology, Vascular and Metabolic Diseases, Department of Internal Medicine, Cardiovascular Research Institute "COEUR", Erasmus MC, University Medical Centre Rotterdam, P.O. Box 2040, 3000 CA Rotterdam, The Netherlands

ARTICLE INFO

*Article history:*
Received 22 October 2008
Received in revised form 16 April 2009
Accepted 29 April 2009
Available online 19 May 2009

*Keywords:*
BRL15572
Capsaicin
α-CGRP
External carotid vasodilatation
Migraine
SB224289
Sumatriptan

ABSTRACT

Migraine is a neurovascular disorder associated with trigeminal activation, vasodilatation and trigeminal release of calcitonin gene-related peptide (CGRP). The antimigraine properties of triptans may be due to: i) vasoconstriction of the carotid arterial bed via 5-$HT_{1B}$ receptors; and ii) inhibition of CGRP release from trigeminal nerves, via 5-$HT_{1B/1D}$ receptors. This study investigated the effects of intrathecally administered sumatriptan (a 5-$HT_{1B/1D}$ receptor agonist) and PNU-142633 (a 5-$HT_{1D}$ receptor agonist) on the canine external carotid vasodilator responses to capsaicin, α-CGRP and acetylcholine. For this purpose, 42 mongrel dogs were anaesthetised with sodium pentobarbitone and, subsequently, vagosympathectomized. The animals were prepared to measure arterial blood pressure, heart rate and external carotid blood flow; the thyroid artery was cannulated for infusion of agonists. 1-min intracarotid (i.c.) continuous infusions of capsaicin, α-CGRP and acetylcholine produced dose-dependent increases in external carotid blood flow without affecting arterial blood pressure or heart rate. These vasodilator responses remained unaffected after intrathecal (i.t.) administration of physiological saline (0.5 ml) or PNU-142633 (300–1000 µg); in contrast, i.t. sumatriptan (300–1000 µg) significantly inhibited the vasodilator responses to capsaicin, but not those to α-CGRP or acetylcholine. Furthermore, i.t. administration of SB224289 (a 5-$HT_{1B}$ receptor antagonist), but not of BRL15572 (a 5-$HT_{1D}$ receptor antagonist), abolished the above inhibition by sumatriptan. These results suggest that sumatriptan-induced inhibition of the external carotid vasodilatation to capsaicin involves a central mechanism mainly mediated by 5-$HT_{1B}$ receptors.



## 1. Introduction

Migraine, a neurovascular disorder associated with activation of perivascular trigeminal sensory nerves (Williamson and Hargreaves, 2001; Peroutka, 2005), consists of three phases: (i) a trigger phase characterized by neuronal hyperexcitability; (ii) an aura phase (before the onset of headache) which seems to involve cortical spreading depression followed by a decrease in regional cerebral blood flow; and (iii) a headache phase probably due to cranial vasodilatation (see Villalón et al., 2002, 2003). Thus, it has been suggested that dilatation of cranial blood vessels involves trigeminal release of calcitonin gene-related peptide (CGRP) (see Arulmani et al., 2004).

Moreover, an intravenous (i.v.) infusion of CGRP can cause a migraine-like headache in the majority of migraine patients (Lassen et al., 2002); while triptans can decrease plasma concentrations of CGRP in parallel with alleviation of headache (see Goadsby et al., 2002; Villalón et al., 2003; Juhasz et al., 2005). In keeping with the above, Muñoz-Islas et al. (2006) have shown in dogs that i.v. donitriptan (a lipid-soluble 5-$HT_{1B/1D}$ receptor agonist), but not sumatriptan (a water-soluble 5-$HT_{1B/1D}$ receptor agonist), PNU-142633 (a water-soluble 5-$HT_{1D}$ receptor agonist) or PNU-109291 (a lipid-soluble 5-$HT_{1D}$ receptor agonist), inhibited the external carotid vasodilator responses to capsaicin (a response mediated via CGRP release). This inhibitory effect of donitriptan, being blocked by i.v. SB224289 (a 5-$HT_{1B}$ receptor antagonist), but not by BRL15572 (a 5-$HT_{1D}$ receptor antagonist), seems to involve stimulation of central 5-$HT_{1B}$ receptors (Muñoz-Islas et al., 2006), as previously reported (Goadsby and Hosking, 1998; Goadsby et al., 2002). The lack of effect of sumatriptan may be explained in terms that this triptan: i) does not easily cross an intact blood–brain barrier (Humphrey et al., 1991); (ii) can cross the blood–brain barrier only after its disruption with a hyperosmolar mannitol solution (Kaube et al., 1993); and (iii) displays a weaker agonist potency at central 5-$HT_{1B/1D}$ receptors compared

* Corresponding author. Tel.: +52 55 5483 2854; fax: +52 55 5483 2863.
*E-mail address:* cvillalon@cinvestav.mx (C.M. Villalón).
*URL:* http://www.cinvestav.mx/farmacobiologia/PersonalAcademico/vhcm.html (C.M. Villalón).

0014-2999/$ – see front matter 
doi:10.1016/j.ejphar.2009.04.070

**Table 1**
Binding affinity constants (p*K*i) of several 5-HT receptor agonists and antagonists for cloned human 5-$HT_1$ receptor subtypes.

| | 5-$HT_{1A}$ | 5-$HT_{1B}$ | 5-$HT_{1D}$ | 5-$ht_{1E}$ | 5-$HT_{1F}$ |
|---|---|---|---|---|---|
| Sumatriptan | 6.4[a] | 7.8[a] | 8.5[a] | 5.8[a] | 7.9[a] |
| PNU-142633 | – | 4.8[b] | 8.3[b] | – | – |
| SB224289 | 5.5[c] | 8.0[c] | 6.2[c] | <5.0[c] | <5.0[c] |
| BRL15572 | 7.7[d] | 6.1[d] | 7.9[d] | 5.2[d] | 6.0[d] |

Data taken from: [a]Leysen et al. (1996); [b]Pregenzer et al. (1999); [c]Hagan et al. (1997); and [d]Price et al. (1997).

with eletriptan and zolmitriptan after i.v. administration (Johnson et al., 2001). Hence, although it is known that a central mechanism (spinal and/or supraspinal) participates in the inhibitory effect of triptans, it is unclear whether the in situ spinal administration of sumatriptan can inhibit the external carotid vasodilator responses to capsaicin.

On this basis, the present study set out to investigate in vagosympathectomized dogs: (i) whether intrathecal (i.t.) administration of either sumatriptan or PNU-142633 is capable of inhibiting the external carotid vasodilator responses to capsaicin, α-CGRP and acetylcholine; and (ii) whether 5-$HT_{1B}$ and/or 5-$HT_{1D}$ receptors are involved in the above responses by using SB224289 (a 5-$HT_{1B}$ receptor antagonist) and BRL15572 (a 5-$HT_{1D}$ receptor antagonist) (see Table 1).

## 2. Materials and methods

### *2.1. General methods*

Experiments were carried out in a total of 42 male mongrel dogs (17–25 kg) that were anaesthetised with sodium pentobarbitone (30 mg/kg, i.v. and, when required, 1 mg/kg for maintaining the anaesthesia; see Villalón et al., 1999). All dogs were artificially respired with room air, using a Palmer ventilation pump (rate: 20 strokes/min; stroke volume: 13–16 ml/kg) as previously established by Kleinman and Radford (1964). A catheter was placed intrathecally (Yaksh et al., 2004), through the subarachnoid space (spinal $C_1$–$C_3$ region) for administration of vehicle (saline or propylene glycol), sumatriptan, PNU-142633, SB224289 or BRL15572. Another catheter was placed in a femoral artery, connected to a blood pressure transducer (P23XL, Grass Instrument Co., Quincy, MA, U.S.A.), for measurement of arterial blood pressure. After administration of vehicle, agonists or antagonists, the i.t. catheter was flushed with 0.3 ml of saline. Mean blood pressure was calculated from the systolic and diastolic arterial pressures as: Mean blood pressure = diastolic pressure + (systolic pressure – diastolic pressure)/3. Heart rate was measured with a tachograph (7P4F, Grass Instrument Co., Quincy, MA, U.S.A.) triggered from the blood pressure signal.

The right common carotid artery was dissected free and the corresponding internal carotid and occipital branches were ligated; under these experimental conditions, the blood flow through the right common carotid artery was considered to represent that of the external carotid artery (for further considerations see Villalón et al., 1999). Thereafter, an ultrasonic flow probe (4 mm R-Series) connected to an ultrasonic T206 flow meter (Transonic Systems Inc., Ithaca, NY, U.S.A.) was placed around the right common carotid artery. Bilateral cervical vagosympathectomy was systematically performed in order to prevent possible baroreceptor reflexes induced by the intracarotid (i.c.) infusions of compounds. Subsequently, a catheter was introduced into the right cranial thyroid artery for the i.c. infusions of capsaicin, α-CGRP and acetylcholine.

It is to be noted that after bilateral cervical vagosympathectomy the carotid arterioles are dilated; since the main objective of our study is to investigate the responses to vasodilator agents (such as capsaicin, α-CGRP and acetylcholine) on the external carotid circulation, a selective carotid preconstriction was induced with an i.c. continuous infusion of phenylephrine (an $\alpha_1$-adrenoceptor agonist) in order to enhance the vasodilator responses to capsaicin, as previously reported (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006). For this purpose, a 0.5 mm (external diameter) needle, connected to a suitable catheter, was inserted into the right common carotid artery for the continuous infusion of phenylephrine by another motor driven syringe. Then, the vasodilator responses to capsaicin during the i.c. continuous infusion of phenylephrine were compared with those observed after i.t. administration of sumatriptan, PNU-142633 or equivalent volumes of physiological saline. By using WPI model sp100i pumps (World Precision Instruments Inc., Sarasota, FL, U.S.A.), capsaicin, α-CGRP, acetylcholine and phenylephrine were infused into the carotid artery (for further details, see the experimental protocols described below).

Arterial blood pressure, heart rate and external carotid blood flow were recorded simultaneously by a model 7D polygraph (Grass Instrument Co., Quincy, MA, U.S.A.). The body temperature of the animals was maintained between 37–38 °C.

### *2.2. Experimental protocols*

After the animals ($n = 42$) had been in a stable haemodynamic condition for at least 60 min, baseline values of mean blood pressure, heart rate and external carotid blood flow were determined. Then, all animals received an i.c. continuous infusion of phenylephrine (1.5 µg/min) to obtain a sustained carotid vasoconstriction. 20 min later all the different treatments were given *without interrupting the infusion of phenylephrine*. Then the animals were divided into five groups ($n = 8$, 6, 7, 11 and 10).

The first group ($n = 8$) received consecutive i.c. infusions (1 ml/min, for 1 min) of capsaicin (10, 18, 31 and 56 µg/min). Then, this group was subdivided into two subgroups ($n = 4$ each) that received i.t. bolus injections (0.5 ml) of, respectively: (i) vehicle (physiological saline; three times); and (ii) sumatriptan (100, 300 and 1000 µg given cumulatively). 10 min after i.t. administration of each dose of sumatriptan or saline (0.5 ml), dose–response curves to capsaicin were elicited; those were completed 30 min after starting the infusion of the first dose of capsaicin. The next dose of sumatriptan or its equivalent volume of saline was given when the administration scheme of capsaicin had been finished.

The second group ($n = 6$) received consecutive i.c. infusions (1 ml/min, for 1 min) of capsaicin (10, 18, 31 and 56 µg/min), α-CGRP (0.1, 0.3, 1 and 3 µg/min) and acetylcholine (0.01, 0.03, 0.1 and 0.3 µg/min). Then, this group was subdivided into two subgroups that received i.t. bolus injections (0.5 ml) of, respectively: (i) vehicle (saline; $n = 3$); or (ii) 1000 µg sumatriptan (which induced the maximum inhibition; $n = 3$). 10 min later, the responses to the above infusions of capsaicin, α-CGRP and acetylcholine were elicited again.

The third group ($n = 7$) received i.c. infusions of capsaicin, α-CGRP and acetylcholine as described above. Then, this group was subdivided into two subgroups that received i.t. bolus injections (0.5 ml) of, respectively: (i) vehicle (saline; two times; $n = 3$); and (ii) PNU-142633 (300 and 1000 µg given cumulatively; $n = 4$). The responses to capsaicin, α-CGRP and acetylcholine (in this order) were elicited again 10 min after the i.t. administration of each dose of PNU-142633 or its equivalent volumes of saline.

The fourth group ($n = 11$) was subdivided into three subgroups that received i.c. infusions of the above doses of capsaicin before and 10 min after i.t. bolus injections of, respectively: (i) vehicle (propylene glycol; 0.5 ml; $n = 4$); (ii) SB224289 (1000 µg; $n = 4$); and (iii) BRL15572 (1000 µg; $n = 3$).

The last group ($n = 10$) received the above infusions of capsaicin. At this point, this group was subdivided into three subgroups that received i.t. bolus injections of, respectively, (i) SB224289 (1000 µg;

**Table 2**
Values of mean arterial blood pressure (MAP; mmHg), heart rate (HR; beats/min) and external carotid blood flow (ECBF; ml/min), before and after: (i) 20 min the i.c. continuous infusion of phenylephrine (1.5 μg/min) had been started; and (ii) 10 min of the i.t. bolus administration of physiological saline (0.5 ml), sumatriptan (1000 μg), PNU-142633 (1000 μg), propylene glycol (0.5 ml), SB224289 (1000 μg) or BRL15572 (1000 μg).

| | | MAP mmHg | | HR beats/min | | ECBF ml/min | |
|---|---|---|---|---|---|---|---|
| | *n* | Before | After | Before | After | Before | After |
| Phenylephrine | 42 | 140 ± 3 | 140 ± 3 | 165 ± 4 | 162 ± 3 | 208 ± 12 | 127 ± 7* |
| Saline | 10 | 140 ± 4 | 156 ± 12 | 161 ± 8 | 154 ± 6 | 128 ± 14 | 129 ± 11 |
| Sumatriptan | 7 | 156 ± 9 | 148 ± 11 | 161 ± 6 | 151 ± 5 | 101 ± 15 | 109 ± 15 |
| PNU-142633 | 4 | 138 ± 7 | 142 ± 7 | 150 ± 7 | 158 ± 1 | 130 ± 11 | 124 ± 19 |
| Propylene glycol | 7 | 139 ± 8 | 137 ± 7 | 168 ± 4 | 156 ± 9 | 116 ± 7 | 114 ± 10 |
| SB224289 | 8 | 134 ± 4 | 137 ± 10 | 170 ± 6 | 169 ± 5 | 119 ± 10 | 111 ± 10 |
| BRL15572 | 6 | 155 ± 8 | 150 ± 8 | 169 ± 4 | 163 ± 3 | 108 ± 12 | 98 ± 12 |

*$P<0.05$ vs baseline value (before).

$n=4$); (ii) BRL15572 (1000 μg; $n=3$); and (iii) an equivalent volume of vehicle (0.5 ml; $n=3$). After 10 min, each subgroup received an i.t. bolus injection of sumatriptan (1000 μg) as previously described and, 10 min later, the responses to the above i.c. infusions of capsaicin were elicited again.

### 2.3. Other procedures applying to all protocols

Each dose of capsaicin, α-CGRP and acetylcholine was in a solution administered at a rate of 1 ml/min during 1 min. The dose-intervals between the different doses of these compounds ranged between 5 min (acetylcholine) and 20 min (capsaicin and α-CGRP), as in each case we waited until the external carotid blood flow had returned completely to baseline values. The different doses were given in a sequential scheme in view that they produced transient responses. The doses of these compounds were selected on the basis of results previously published (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006), in which dose-dependent increases in external carotid blood flow were elicited with no changes in blood pressure or heart rate.

The correctness of the i.t. administration of the different compounds was validated at the end of all experiments (after sodium pentobarbitone overdose) by injecting 1% methylene blue (0.5 ml, i.t.); then, the spinal cord was dissected by laminectomy at the level of the $C_1$–$C_4$ intervertebral disc to look for the catheter tip and dye distribution at this level. The dogs showing the catheter tip positioned at sites other than the $C_1$–$C_3$ spinal cord (only 2 dogs) were not considered for data analysis.

The Ethical Committee of our institution (CICUAL), dealing with the use of animals in scientific experiments, approved the protocols of the present investigation.

### 2.4. Drugs

Apart from the anaesthetic (sodium pentobarbitone), the compounds used in the present study (obtained from the sources indicated) were: L-phenylephrine hydrochloride, capsaicin, rat α-CGRP, acetylcholine chloride and SB224289 (2,3,6,7-tetrahydro-1'-methyl-5-[2'-methyl-1,2,4-oxadiazol-3-yl)biphenyl-4-carbonyl] furo [2,3f] indole-3-spiro-4'-piperidine hydrochloride) (Sigma Chemical Co., St. Louis, MO, U.S.A.); sumatriptan succinate (gift from Dr. Paul J. Strijbos, GlaxoSmithKline, Harlow, U.K.); BRL15572 (1-(3chorophenyl)4-[3,3-diphenyl (2-(S,R)hydroxy propanyl) piperazine] hydrochoride) (Tocris Cookson Inc. Ellisville, MO, U.S.A.) and PNU-142633 [(S)-(-)-3,4-dihydro-1-[2-[4-aminocarbonyl)phenyl]-1-piperazinyl] ethyl]-N-methyl-1H-2-benzopyran-6-carboxamide] (gift from Dr. Robert B. McCall, Pharmacia and Upjohn, Kalamazoo, MI, U.S.A.). All compounds were dissolved in physiological saline. When needed, some drops of 20% (v/v) propylene glycol (SB224289 and BRL15572) or 20% (v/v) ethanol (capsaicin) were added; and, then, the resulting solution was finally diluted with physiological saline; and fresh solutions were prepared for each experiment. These vehicles had no effect (when given i.t. or i.c.) on the external carotid blood flow, blood pressure or heart rate. The doses of the antagonists refer to their respective salts, whilst those of the agonists refer to their free base.

### 2.5. Data presentation and statistical evaluation

All data are presented as mean ± S.E.M. The peak changes in external carotid blood flow were expressed as percent change from baseline. The difference between the variables within one group (or *different groups) of animals was compared by using a two-way repeated measures analysis of variance (or, correspondingly, *a two-way analysis of variance) (both randomized block design) followed by the Student–Newman–Keuls' test (Steel and Torrie, 1980). Statistical significance was accepted at $P<0.05$.

## 3. Results

### 3.1. Systemic and carotid haemodynamic effects of the different treatments

Baseline values of mean blood pressure, heart rate and external carotid blood flow in the 42 anaesthetised dogs were 140 ± 3 mmHg, 165 ± 4 beats/min and 208 ± 12 ml/min, respectively. A significant (and sustained) decrease in external carotid blood flow (38.9%) was induced 20 min after the i.c. continuous infusion of phenylephrine had been started, without significant changes in mean blood pressure or heart rate (see Table 2) ($n=42$). Under these experimental conditions, the above variables were not significantly modified in the animals that received the i.t. bolus injection of physiological saline (0.5 ml; $n=10$); sumatriptan (1000 μg; $n=7$); PNU-142633 (1000 μg; $n=4$); propylene glycol (0.5 ml; $n=7$); SB224289 (1000 μg; $n=8$) or BRL15572 (1000 μg; $n=6$) (see Table 2).

### 3.2. Effect of increasing doses of sumatriptan (given i.t.) on the external carotid vasodilator responses to capsaicin

1-min i.c. infusions of capsaicin induced dose-dependent increases in the external carotid blood flow (see Fig. 1A; control response); these effects were not accompanied by changes in mean blood pressure or heart rate (not shown), as previously reported (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006). These vasodilator responses to capsaicin, particularly those induced by 30 and 56 μg/min: (i) were significantly inhibited after i.t. administration of 1000 μg (but not 100 μg) sumatriptan (Fig. 1B); and (ii) remained without significant changes after i.t. administration of the corresponding

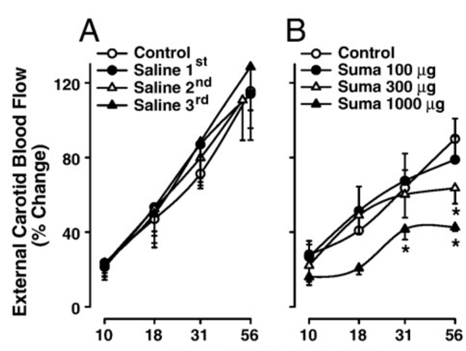


**Fig. 1.** Effect of 1-min intracarotid (i.c.) continuous infusions of capsaicin on the canine external carotid blood flow before (control responses) and after intrathecal (i.t.) bolus injections of either: (A) physiological saline (3 times; $n=4$); or (B) sumatriptan (Suma; 100–1000 μg; $n=4$). *$P<0.05$ vs. the corresponding control response. Note that the control responses in B did not significantly differ from those in A ($P>0.05$).

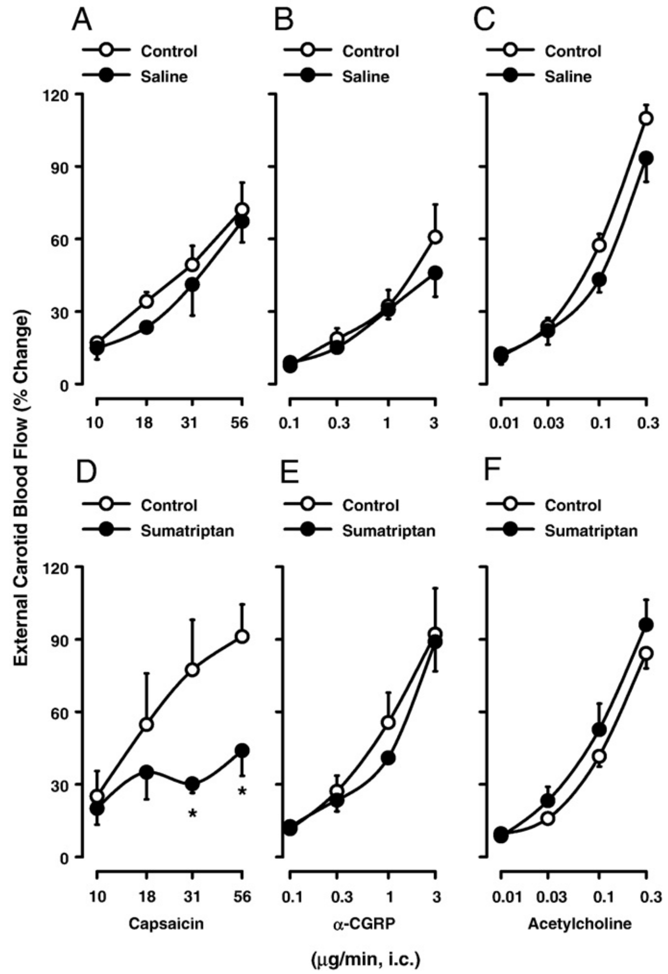


**Fig. 2.** Effect of 1-min intracarotid (i.c.) continuous infusions of capsaicin, α-CGRP or acetylcholine on the canine external carotid blood flow before (control responses) and after an intrathecal (i.t.) bolus injection of either: (i) (A, B and C) 0.5 ml vehicle (physiological saline, $n=3$); or (ii) (D, E and F) 1000 μg sumatriptan ($n=3$). *$P<0.05$ vs. the corresponding control response.

volumes of vehicle (0.5 ml of physiological saline given three times; see Fig. 1A). Note that the control responses to capsaicin in the groups treated with saline (Fig. 1A) or sumatriptan (Fig. 1B) did not significantly differ ($P>0.05$).

### *3.3. Effect of an i.t. bolus injection of saline, sumatriptan or PNU-142633 on the external carotid vasodilator responses to capsaicin, α-CGRP and acetylcholine*

1-min i.c. infusions of capsaicin, α-CGRP and acetylcholine induced dose-dependent increases in the external carotid blood flow (see Fig. 2; control responses) without changes in mean blood pressure or heart rate (not shown), as previously reported (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006). These vasodilator responses to capsaicin and α-CGRP were longer-lasting (between 5 and 20 min) than those to acetylcholine (between 1 and 5 min). Moreover, the vasodilator responses to the three compounds remained unchanged after i.t. bolus injections of: (i) vehicle (0.5 ml of physiological saline) given once (see Fig. 2A,B and C) or twice (Fig. 3A,B and C); or (ii) PNU-142633 (300 and 1000 μg; Fig. 3D,E and F).

Furthermore, the external carotid vasodilator responses to capsaicin (particularly at 31 and 56 μg/min) were significantly inhibited after i.t. sumatriptan (1000 μg; Fig. 2D), whereas the responses to α-CGRP (Fig. 2E) and acetylcholine (Fig. 2F) remained without significant changes.

### *3.4. Effect of i.t. bolus injections of vehicle, SB224289 or BRL15572 on the external carotid vasodilator responses to capsaicin and on sumatriptan-induced inhibition of the external carotid vasodilator responses to capsaicin*

As shown in Fig. 4 (upper panel), i.t. administration of SB224289 (1000 μg; Fig. 4B), BRL15572 (1000 μg; Fig. 4C) or equivalent volumes of vehicle (0.5 ml propylene glycol; Fig. 4A) failed to significantly modify *per se* the vasodilator responses to capsaicin. Then, we considered it appropriate to identify the pharmacological profile of sumatriptan-induced inhibition of the external carotid vasodilator responses to capsaicin by analyzing the effects produced by i.t. administration of the above compounds. Hence, Fig. 4 (lower panel) demonstrates that the inhibition exerted by sumatriptan on capsaicin-induced external carotid vasodilator responses was: (i) completely antagonized by 1000 μg SB224289 (Fig. 4E); and (ii) unaffected by 1000 μg BRL15572 (Fig. 4F) or equivalent volumes of vehicle (0.5 ml propylene glycol; Fig. 4D). In fact, when comparing the inhibitory effects of sumatriptan after BRL15572 (Fig. 4F) or vehicle (Fig. 4D), there was no significant difference ($P>0.05$).

## 4. Discussion

### *4.1. General*

Apart from the implications discussed below, our results demonstrate that: (i) i.c. infusions of capsaicin, α-CGRP and acetylcholine can

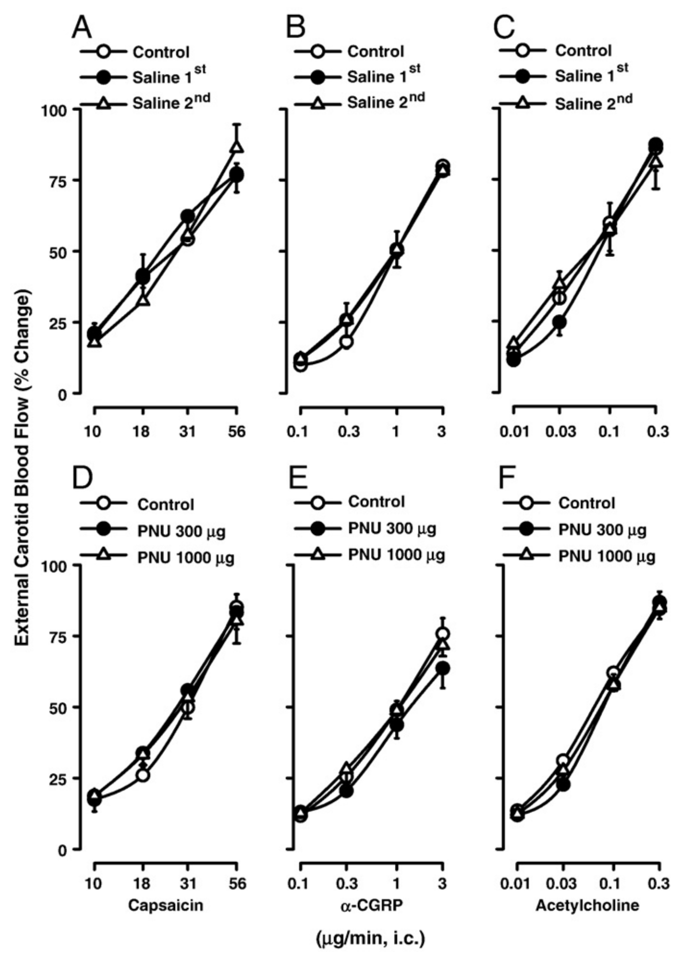


**Fig. 3.** Effect of 1-min intracarotid (i.c.) continuous infusions of capsaicin, α-CGRP and acetylcholine on the canine external carotid blood flow before (control responses) and after intrathecal (i.t.) bolus injections of either: (i) (A, B and C) 0.5 ml vehicle (physiological saline given twice, $n=3$); or (ii) (D, E and F) PNU-142633 (PNU; 300 and 1000 μg; $n=4$). The corresponding i.t. injections of vehicle and PNU-142633 were devoid of any significant effect ($P>0.05$).

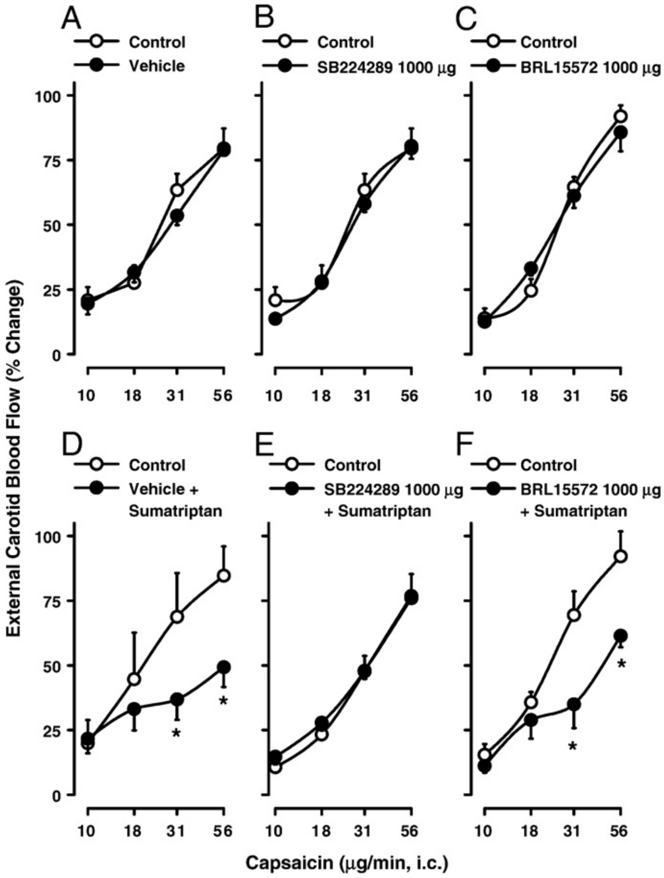


**Fig. 4.** Effect of intrathecal (i.t.) bolus injections of vehicle (0.5 ml propylene glycol), SB224289 (1000 µg) or BRL15572 (1000 µg) on: (i) the external carotid vasodilator responses induced by 1-min intracarotid (i.c.) continuous infusions of capsaicin (A, B and C; $n=4$, 4 and 3, respectively); and (ii) sumatriptan (1000 µg)-induced inhibition of the external carotid vasodilator responses to capsaicin (D, E and F; $n=3$, 4 and 3, respectively). $^{*}P<0.05$ vs. the corresponding control response.

induce dose-dependent (and highly reproducible) external carotid vasodilator responses in the dog (Figs. 1 and 2), as previously shown (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006); (ii) i.t. administration of sumatriptan (Fig. 2), but not of PNU-142633 (Fig. 3), can specifically inhibit capsaicin-induced external carotid vasodilatation (unlike i.v. sumatriptan; Muñoz-Islas et al., 2006), as the responses to α-CGRP and acetylcholine remained unaltered; and (iii) i.t. pretreatment with SB224289, but not with BRL15572, abolished the above inhibition by sumatriptan (Fig. 4), reinforcing the involvement of a central mechanism mediated by 5-$HT_{1B}$ (rather than 5-$HT_{1D}$) receptors.

### *4.2. Systemic and carotid haemodynamic changes by the different treatments*

It is important to note that i.t. administration of vehicles (saline or propylene glycol), SB224289, BRL15572, sumatriptan and PNU-142633, as well as the i.c. continuous infusions of saline, did not affect mean blood pressure, heart rate and external carotid blood flow (see Results). Therefore, no time-dependent changes occurred in the haemodynamic variables during the course of our experimental protocols. Since the i.c. infusion of phenylephrine decreased the external carotid blood flow (about 38.9%) without changing blood pressure or heart rate, a local vasoconstrictor effect mediated by $\alpha_1$-adrenoceptors is implied, as previously shown by Willems et al. (2001). Moreover, although it could be argued that the inhibition by sumatriptan could have been due to leakage into the circulation, this possibility is ruled out because no significant decreases in external carotid blood flow were observed after spinal administration of sumatriptan (100–1000 µg) (see Table 2).

### *4.3. Mechanisms involved in the external carotid vasodilator responses to capsaicin, α-CGRP and acetylcholine*

Although not specifically characterized in the present study, previous lines of evidence in the carotid circulation suggest that the external carotid vasodilator responses to capsaicin, α-CGRP and acetylcholine in our experiments may involve: (i) CGRP release and activation of BIBN4096BS (a CGRP receptor antagonist)-sensitive CGRP receptors for capsaicin (Kapoor et al., 2003a); (ii) BIBN4096BS-sensitive CGRP receptors for α-CGRP (Kapoor et al., 2003b); and (iii) atropine-sensitive muscarinic receptors located on the vascular endothelium for acetylcholine (Villalón et al., 1993).

### *4.4. Specific inhibition by spinal sumatriptan on the vasodilator responses to capsaicin*

As previously shown, i.c. phenylephrine decreases the external carotid blood flow and enhances the vasodilator responses to capsaicin, α-CGRP and acetylcholine in a dose- dependent and reproducible manner (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006). For that reason, we decided to perform our experimental protocols in the present study during an i.c. continuous infusion of phenylephrine; under these experimental conditions, the i.t. administration of sumatriptan (and of the rest of compounds; see Results) induced no significant changes in external carotid blood flow, blood pressure and heart rate as the doses required to interact with their respective receptors are much lower (present results; 300 and 1000 µg) than those given via the i.v. route (300 µg/kg) (Muñoz-Islas et al., 2006). So, any effect of sumatriptan on the external carotid vasodilator responses to capsaicin, α-CGRP or acetylcholine should be attributed to a direct interaction of this triptan with their respective receptors (5-$HT_{1B/1D}$). Hence, the fact that spinal sumatriptan (1000 µg) inhibited the vasodilator responses to capsaicin without affecting those to α-CGRP and acetylcholine (Fig. 2), demonstrates that such an inhibition: (i) is highly specific; (ii) involves inhibition of trigeminal neurotransmission rather than a postjunctional interaction (on vascular CGRP or muscarinic receptors); and (iii) involves a central (spinal), rather than a peripheral mechanism (see below).

In addition, the inhibitory effect produced by spinal sumatriptan suggests that not only the supraespinal level (Lambert et al., 1992; Ellrich et al., 2001), but also that the spinal level ($C_1$–$C_3$) may play an important role in its antimigraine action. The latter is consistent with the fact that spinal sumatriptan inhibits the paw edema to subcutaneous carragenan (Daher et al., 2005), a finding that reinforces the involvement of a spinal mechanism in the modulation of the neurogenic inflammation induced by a peripheral chemical stimulus.

### *4.5. Involvement of 5-$HT_{1B}$, rather than 5-$HT_{1D}$, receptors in the inhibition by i.t. sumatriptan*

It has previously been shown that activation of 5-$HT_{1B}$, 5-$HT_{1D}$ and 5-$HT_{1F}$ receptors (in decreasing order of potency) inhibits the trigeminovascular nociceptive transmission in cats (Goadsby and Classey, 2003). Our study in dogs is in keeping with the above findings only regarding the role of 5-$HT_{1B}$ receptors (probably due to differences in species and/or experimental conditions), since the inhibition by i.t. sumatriptan on capsaicin-induced external carotid vasodilatation was: (i) not mimicked by i.t. PNU-142633 (up to 1000 µg; Fig. 3D), a selective 5-$HT_{1D}$ receptor agonist (see Table 1); and (ii) abolished by i.t. SB224289 (Fig. 4E), but unaffected by i.t. BRL15572 (Fig. 4F), at doses (1000 µg) that did not affect *per se* the vasodilator responses to capsaicin (Fig. 4B and C, respectively).

Although sumatriptan displays a reasonable affinity for 5-$HT_{1F}$ and a lower affinity for 5-$HT_{1A}$ receptors (see Table 1), the involvement of both receptors seems unlikely since: (i) SB224289 (which abolished the inhibition by sumatriptan) has a low affinity for these receptors

(Table 1); and (ii) BRL15572 (which did not affect the inhibition by sumatriptan) displays a reasonable affinity for these receptors (Table 1). Admittedly, our suggestion supporting the role of sumatriptan-sensitive 5-$HT_{1B}$ receptors is based on the assumption that species differences between the binding of sumatriptan, PNU-142633, SB224289 and BRL15572 to canine and human 5-$HT_{1B}$ and 5-$HT_{1D}$ receptors do not play a major role (see Table 1).

### *4.6. Possible locus of the 5-$HT_{1B}$ receptors involved in the inhibitory action of i.t. sumatriptan*

The fact that i.t. sumatriptan in dogs (at the spinal $C_1$–$C_3$ region) inhibited the vasodilator responses to capsaicin suggests the involvement of a spinal mechanism. The latter is consistent with other findings using zolmitriptan (a lipid-soluble 5-$HT_{1B/1D}$ receptor agonist) in cats showing: (i) the existence of receptors that specifically bind this triptan on the superficial laminae of the trigeminal nucleus caudalis (Goadsby and Knight, 1997); and (ii) a reduction in both c-*Fos* expression at the medulla dorsal horns (trigeminal nucleus caudalis) and second-order trigeminal neuronal activity (Hoskin and Goadsby, 1998). Therefore, central trigeminal sites could be involved in the processing of craniovascular pain. In agreement with the above findings, the most likely site of action for brain penetrating triptans, as previously suggested for i.v. donitriptan (Muñoz-Islas et al., 2006), may be the first-order trigeminal nucleus synapse in the brainstem and upper cervical cord. In this site of the brainstem some kind of effect on antidromic transmission and/or alterations in some reflex pathways passing through the central nervous system might be involved, as suggested for sumatriptan in the spinal control of peripheral inflammation in the rat (Daher et al., 2005). However, the latter is a mere speculation and additional experiments, which fall beyond the scope of the present study, will be required to further investigate the involvement of these reflexes in the inhibitory effect of spinal sumatriptan on the vasodilator responses to capsaicin.

In conclusion, the above results, taken together, suggest that the inhibition produced by i.t. sumatriptan on capsaicin-induced external carotid vasodilatation is mainly mediated by central 5-$HT_{1B}$ receptors probably located in the trigeminocervical complex.

## Acknowledgements

The authors would like to thank Mr. Arturo Contreras for his skilful technical assistance. We are also indebted to the pharmaceutical companies (see Drugs section) and CONACyT (Mexico) for their support.

## References

Arulmani, U., MaassenVanDenBrink, A., Villalón, C.M., Saxena, P.R., 2004. Calcitonin gene-related peptide and its role in migraine pathophysiology. Eur. J. Pharmacol. 500, 315–330.

Daher, J.B., de Melo, M.D., Tonussi, C.R., 2005. Evidence for a spinal serotonergic control of the peripheral inflammation in the rat. Life Sci. 76, 2349–2359.

Ellrich, J., Messlinger, K., Chiang, C.Y., Hu, J.W., 2001. Modulation of neuronal activity in the nucleus raphé magnus by the 5-HT1-receptor agonist naratriptan in rat. Pain 90, 227–231.

Goadsby, P.J., Classey, J.D., 2003. Evidence for serotonin $(5\text{-HT})_{1B}$, 5-$HT_{1D}$ and 5-$HT_{1F}$ receptor inhibitory effects on trigeminal neurons with craniovascular input. Neuroscience 122, 491–498.

Goadsby, P.J., Knight, Y.E., 1997. Direct evidence for central sites of action of zolmitriptan (311C90): an autoradiographic study in cat. Cephalalgia 17, 153–158.

Goadsby, P.J., Hosking, K.L., 1998. Serotonin inhibits trigeminal nucleus activity evoked by craniovascular stimulation through a 5-$HT_{1B/1D}$ receptor: a central action in migraine? Ann. Neurol. 43, 711–718.

Goadsby, P.J., Lipton, R.B., Ferrarri, M.D., 2002. Migraine-current understanding and treatment. N. Engl. J. Med. 346, 257–270.

Hagan, J.J., Slade, P.D., Gaster, L., Jeffrey, P., Hatcher, J.P., Middlemiss, D.N., 1997. Stimulation of 5-$HT_{1B}$ receptors causes hypothermia in the guinea pig. Eur. J. Pharmacol. 331, 169–174.

Hoskin, K.L., Goadsby, P.J., 1998. Comparison of more and less lipophilic serotonin (5.$HT_{1B/1D}$) agonists in a model of trigeminovascular nociception in cat. Exp. Neurol. 150, 45–51.

Humphrey, P.P.A., Feniuk, W., Marriott, A.S., Tanner, R.J., Jackson, M.R., Tucker, M.L., 1991. Preclinical studies on the anti-migraine drug, sumatriptan. Eur. Neurol. 31, 282–290.

Jiménez-Mena, L.R., Gupta, S., Muñoz-Islas, E., Lozano-Cuenca, J., Sánchez-López, A., Centurión, D., Mehrotra, S., MaassenVanDenBrink, A., Villalón, C.M., 2006. Clonidine inhibits the canine external carotid vasodilatation to capsaicin by $\alpha_{2A/2C}$-adrenoceptors. Eur. J. Pharmacol. 543, 68–76.

Johnson, D.E., Rollema, H., Schmidt, A.W., McHarg, A.D., 2001. Serotonergic effects and extracellular brain levels of eletriptan, zolmitriptan and sumatriptan in rat brain. Eur. J. Pharmacol. 425, 203–210.

Juhasz, G., Zsombok, T., Jakab, B., Nemeth, J., Szolcsanyi, J., Bagdy, G., 2005. Sumatriptan causes parallel decrease in plasma calcitonina gene-related peptide (CGRP) concentration and migraine headache during nitroglycerin induced migraine attack. Cephalalgia 25, 179–183.

Kapoor, K., Arulmani, U., Heiligers, J.P.C., Garrelds, I.M., Willems, E.W., Doods, H., Villalón, C.M., Saxena, P.R., 2003a. Effects of the CGRP receptor antagonist BIBN4096BS on capsaicin-induced carotid haemodynamic changes in anaesthetised pigs. Br. J. Pharmacol. 140, 329–338.

Kapoor, K., Arulmani, U., Heiligers, J.P.C., Willems, E.W., Doods, H., Villalón, C.M., Saxena, P.R., 2003b. Effects of BIBN4096BS on cardiac output distribution and on CGRP-induced carotid haemodynamic responses in the pig. Eur. J. Pharmacol. 475, 69–77.

Kaube, H., Hoskin, K.L., Goadsby, P.J., 1993. Inhibition by sumatriptan of central trigeminal neurons only after blood–brain barrier disruption. Br. J. Pharmacol. 109, 788–792.

Kleinman, L.I., Radford, E.P., 1964. Ventilation standards for small mammals. J. Appl. Physiol. 19, 360–362.

Lambert, G.A., Lowy, A.J., Boers, P.M., Angus-Leppan, H., Zagami, A.S., 1992. The spinal cord processing of input from the superior sagittal sinus: pathway and modulation by ergot alkaloids. Brain Res. 597, 321–330.

Lassen, L.H., Haderslev, P.A., Jacobsen, V.B., Iversen, H.K., Sperling, B., Olesen, J., 2002. CGRP may play a causative role in migraine. Cephalalgia 22, 54–61.

Leysen, J.E., Gommeren, W., Heylen, L., Luyten, W.H., Van de Weyer, I., Vanhoenacker, P., Haegeman, G., Schotte, A., Van Gompel, P., Wouters, R., Lesage, A.S., 1996. Alniditan, a new 5-$hydroxytryptamine_{1D}$ agonist and migraine-abortive agent: ligand-binding properties of human 5-$hydroxytryptamine_{1D\beta}$, human 5-$hydroxytryptamine_{1D\beta}$, and calf 5-$hydroxytryptamine_{1D}$ receptors investigated with [$^3$H]5-hydroxytryptamine and [$^3$H] alniditan. Mol. Pharmacol. 50, 1567–1580.

Muñoz-Islas, E., Gupta, S., Jiménez-Mena, L.R., Lozano-Cuenca, J., Sánchez-López, A., Centurión, D., Mehrotra, S., MaassenVanDenBrink, A., Villalón, C.M., 2006. Donitriptan, but not sumatriptan, inhibits capsaicin-induced canine external carotid vasodilatation via 5-$HT_{1B}$ rather than 5-$HT_{1D}$ receptors. Br. J. Pharmacol. 149, 82–91.

Peroutka, S.J., 2005. Neurogenic inflammation and migraine: implications for the therapeutics. Mol. Interv. 5, 304–311.

Pregenzer, J.F., Alberts, G.L., Bin Im, W., Slightom, J.L., Ennis, M.D., Hoffman, R.L., Ghazal, N.B., TenBrink, R.E., 1999. Differential pharmacology between the guinea-pig and the gorilla 5-$HT_{1D}$ receptor as probed with isochromans (5-$HT_{1D}$-selective ligands). Br. J. Pharmacol. 127, 468–472.

Price, G.W., Burton, M.J., Collin, L.J., Duckworth, M., Gaster, L., Göthert, M., Jones, B.J., Roberts, C., Watson, J.M., Middlemiss, D.N., 1997. SB-216641 and BRL-15572-compounds to pharmacologically discriminate h5-$HT_{1B}$ and h5-$HT_{1D}$ receptors. Naunyn-Schmiedeberg's Arch. Pharmacol. 356, 312–320.

Steel, R.G.D., Torrie, J.H., 1980. Principles and procedures of statistics. A biomedical approach. InMcGraw Hill Kogakusha Ltd., Tokyo.

Villalón, C.M., Terrón, J.A., Hong, E., 1993. Role of 5-$HT_1$-like receptors in the increase in external carotid blood flow induced by 5-hydroxytryptamine in the dog. Eur. J. Pharmacol. 240, 9–20.

Villalón, C.M., De Vries, P., Rabelo, G., Centurión, D., Sánchez-López, A., Saxena, P.R., 1999. Canine external carotid vasoconstriction to methysergide, ergotamine and dihydroergotamine: role of 5-$HT_{1B/1D}$ receptors and $\alpha_2$-adrenoceptors. Br. J. Pharmacol. 126, 585–594.

Villalón, C.M., Centurión, D., Valdivia, L.F., de Vries, P., Saxena, P.R., 2002. An introduction to migraine: from ancient treatment to functional pharmacology and antimigraine therapy. Proc. West. Pharmacol. Soc. 45, 199–210.

Villalón, C.M., Centurión, D., Valdivia, L.F., de Vries, P., Saxena, P.R., 2003. Migraine: an updated review on pathophysiology, pharmacology, therapy and future trends. Curr. Vasc. Pharmacol. 1, 71–84.

Willems, E.W., Valdivia, L.F., Saxena, P.R., Villalón, C.M., 2001. The role of several $\alpha_1$- and $\alpha_2$-adrenoceptor subtypes mediating vasoconstriction in the canine external carotid circulation. Br. J. Pharmacol. 132, 1292–1298.

Williamson, D.J., Hargreaves, R.J., 2001. Neurogenic inflammation in the context of migraine. Microsc. Res. Tech. 53, 167–178.

Yaksh, T.L., Horais, K.A., Tozier, N., Rathbun, M., Richter, P., Rossi, S., Grafe, M., Tong, C., Meschter, C., Cline, J.M., Eisenach, J., 2004. Intrathecal ketorolac in dogs and rats. Toxicol. Sci. 80, 322–334.